Performance Evaluation of Exploration Algorithms for Search-and-Rescue Robots

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Abstract

Search-and-rescue (SAR) robots must rapidly explore cluttered and unknown indoor environments to locate victims under severe time constraints. This project evaluates two widely used exploration strategies: frontier-based exploration and deep reinforcement learning (RL), both within a unified 2D occupancy-grid simulation. The frontier-based approach implements occupancy mapping, Wavefront Frontier Detection, and A* path planning, while RL agents (DQN and PPO) are trained using Stable-Baselines3 on procedurally generated maps. A quantitative comparison is performed across multiple metrics including time-to-find, coverage efficiency, path length, robustness to noise and dynamic changes, and generalization to unseen floorplans.

Experimental results demonstrate a significant generalization gap in learning-based approaches. While RL agents exhibit highly efficient behavior in environments similar to their training distribution, they suffer a sharp drop in performance on unseen maps, with success rates falling to less than 25%. In contrast, the frontier-based method proved superior in robustness, achieving a 100% success rate across all tested scenarios.

Overall, the study reveals a distinct trade-off between peak efficiency and reliability: while reinforcement learning can achieve rapid detections in favorable cases, classical frontier-based exploration remains the only consistent choice for safety-critical SAR applications where failure is not an option.

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1 Introduction

Autonomous exploration is a core capability for mobile robots deployed in search-and-rescue (SAR) missions, where incomplete knowledge of the environment and strict time constraints make manual teleoperation insufficient. Field reports from real disaster responses emphasize that the main challenges robots face are unfamiliar indoor layouts, degraded visibility, and unpredictable structural changes, while still producing timely situational awareness for human teams [1, 2]. These operational demands highlight the need for exploration strategies that remain effective even when prior information about the environment is limited or absent.

A variety of methods have been proposed to guide robot movement through unknown space, ranging from classical geometry-driven techniques to modern learning-based policies. Frontier-based methods, originating from Yamauchi's formulation [3] and later refined for efficiency and scalability [4], provide a structured way for robots to expand their map based on spatial boundaries. More recent reinforcement learning (RL) approaches seek to replace hand-designed heuristics with policies learned through training, exploiting the latest advances in deep learning methods such as DQN [5], policy-gradient algorithms such as PPO [6], and architectures that integrate mapping or graph reasoning [7, 8]. Although RL can discover optimal behaviors, studies consistently report significant challenges when transferring agents to real-world scenarios that differ from the training environments [9, 10].

This project performs a comparative study of two exploration paradigms in a unified 2D occupancy-grid simulator: (i) a classical geometry-driven frontier-based pipeline and (ii) learned policies obtained with deep reinforcement learning (DQN and PPO). The overarching goal is to understand how these two paradigms behave when exposed to both familiar and unfamiliar conditions common in SAR contexts. By evaluating their strengths and weaknesses side-by-side, the study aims to provide insight into which approaches are more suitable for reliable deployment in safety-critical exploration tasks. To keep the scope achievable and scientifically rigorous, we focus primarily on time-to-find while also evaluating coverage, path length, and success rate to support interpretation of performance differences.

1.1 Theoretical Framework and Literature Study

Autonomous exploration typically relies on occupancy grid mapping, where the robot maintains a probabilistic representation of the environment and continually updates it through sensor observations [11]. Exploration is then framed as the problem of selecting actions that reduce map uncertainty while ensuring safe navigation. Classical methods address this through explicit geometric reasoning. The most established approach is frontier-based exploration, introduced by Yamauchi [3], which identifies frontiers – transitions between known and unknown space – as informative exploration targets. Subsequent refinements such as Wavefront Frontier Detection (WFD) [4] and information-theoretic planning methods [12, 13] improve computational efficiency and decision quality by estimating potential information gain.

Learning-based exploration offers an alternative paradigm. Deep reinforcement learning (RL) methods such as DQN [5] and PPO [6] have demonstrated the ability to learn navigation and exploration behaviors directly from interaction. Recent work integrates RL with spatial memory or SLAM components (e.g., Active Neural SLAM [7]) or formulates exploration as a graph-based decision problem [8]. These methods can discover nontrivial strategies but often struggle with generalization, as shown in empirical studies evaluating transfer to unseen layouts or noisy conditions [9, 10]. This limitation is especially relevant for search-and-rescue scenarios, where robots must operate reliably in unfamiliar, unstructured environments without the guarantee that test conditions resemble training distributions.

Given these considerations, classical frontier-based exploration and RL-based exploration represent two contrasting theoretical approaches: one grounded in deterministic geometric reasoning, the other in data-driven policy learning. Their complementary strengths provide the rationale for a controlled comparison within this project.

1.2 Research Questions and Hypotheses

In this project we investigate the performance of two different exploration strategies: frontier-based exploration and reinforcement learning-based exploration within the context of indoor search-and-rescue (SAR) operations. The primary objective is to understand how these approaches differ in terms of efficiency, robustness, and generalization when operating under limited prior knowledge and time-critical constraints. To enable a systematic comparison, the study focuses on quantitative performance measures that are directly relevant to SAR scenarios.

Throughout the evaluation, time-to-find is defined as the number of simulator timesteps required for the robot to first detect the victim. Success rate measures the proportion of runs in which the victim is found within a fixed time budget. Coverage efficiency represents the fraction of free space explored at the time of detection, while path length corresponds to the cumulative distance traveled by the robot until the victim is found. All reported metrics are summarized using the median and inter-quartile range (IQR) to account for skewed distributions.

Based on these definitions, this study addresses the following research questions:

- **RQ1:** In a 2D indoor SAR simulation, which exploration method achieves a lower median time-to-find under nominal sensor noise and map complexity?
- **RQ2:** Which exploration method achieves the highest coverage efficiency, given a fixed amount of time?
- **RQ3:** Which algorithm produces the shortest path to reach the victim?
- **RQ4:** How do these methods differ when evaluated on unseen or structurally different maps?

To guide the experimental evaluation, a set of falsifiable hypotheses is formulated. These hypotheses specify the expected performance trends of the evaluated methods under the experimental conditions considered in this study:

- **H1:** RL agents (DQN, PPO) will achieve lower time-to-find on environments that resemble their training distribution.
- **H2:** Frontier-based exploration will outperform RL agents (i.e., a lower time-to-find) on unseen or structurally different maps due to its geometry-driven decision process.
- **H3:** Under moderate sensor noise and dynamic changes, frontier-based exploration will demonstrate more stable performance, reflected in consistent time-to-find and lower variance across runs
- **H4:** RL agents may discover shorter local paths in familiar environments, but will exhibit a reduced coverage efficiency when generalization is required.

These research questions and hypotheses define the scope of the experimental evaluation and establish clear criteria for interpreting the results presented in later sections of the report.

1.3 Relevance and social impact

This research provides critical insights for system designers and engineers developing autonomous Search-and-Rescue (SAR) robotic platforms. In disaster scenarios, where communication is often cut off and environments are unstructured, choosing the right exploration algorithm is a life-critical decision.
In practice, our comparative analysis serves mainly to pinpoint the central areas of focus for future researchers: by isolating the failure modes of standard Reinforcement Learning (specifically the lack of memory in reactive policies), we identify the precise architectural bottlenecks that must be addressed to make Deep RL viable for real-world SAR missions.

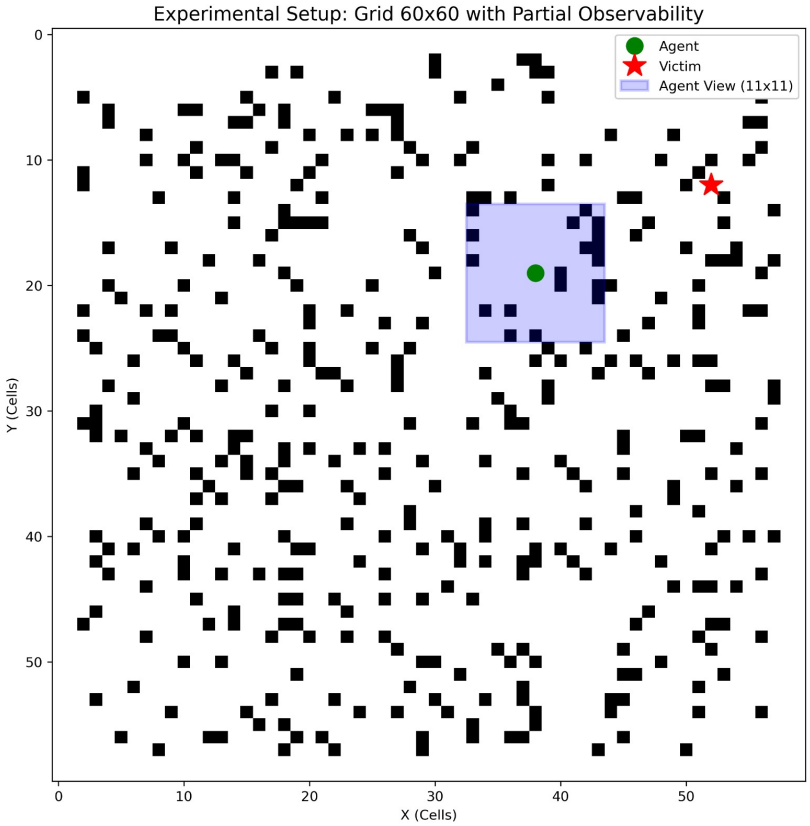


Figure 1: Experimental setup in the 2D grid-world environment. The agent operates in a 60×60 occupancy grid with partial observability, perceiving a local 11×11 observation window centered on its current position. Black cells represent obstacles, the green circle denotes the agent, and the red star indicates the victim location.

2 Method

We used a quantitative experimental methodology to compare a classical frontier-based exploration algorithm and two learned policies (DQN, PPO). All experiments were executed in a controlled 2D occupancy-grid simulator so both methods share identical perception, motion and logging. Next, we describe the environment, algorithms, experimental protocol, metrics and statistical analysis used in the study.

2.1 Simulation environment

Experiments used a grid-world simulator. Main parameters: grid size 60×60 cells, episode length $T_{\max} = 500$ timesteps, LiDAR-like sensor with 36 rays and max range 12 cells, Gaussian sensor noise $\sigma = 0.2$, and slip probability $p_{\text{slip}} = 0.02$. The agent receives a flattened vector representation of a local 11×11 observation window centered on its current position, augmented with the agent's heading and a normalized timestep. Maps were generated procedurally (obstacle density range 0.05–0.15); held-out maps and real floorplans were used for generalization tests. Obstacle density was randomized for each episode to expose agents to diverse spatial layouts and reduce overfitting to specific map structures.

Figure 1 illustrates the experimental setup used throughout this study. The agent operates in a partially observable environment and perceives only a local 11×11 window centered on its current position, while the remainder of the map remains unknown until explored. Obstacles are randomly distributed within the grid, and the victim is placed at a fixed but initially unknown location. This setup reflects the limited sensing conditions typical of indoor search-and-rescue scenarios.

2.2 Algorithms

The exploration strategies evaluated in this study consist of one classical baseline and two learning-based methods. All algorithms operate under identical perception, motion, and termination conditions, ensuring that observed performance differences arise from the exploration strategy itself rather than from differences in sensing or control.

Frontier baseline

The frontier-based baseline follows a classical occupancy-grid exploration pipeline. At each planning cycle, the robot maintains an updated occupancy map constructed from range sensor observations. Frontier cells—defined as boundaries between known free space and unexplored regions—are detected using Wavefront Frontier Detection (WFD). Each detected frontier is scored based on a weighted combination of distance and frontier size, favoring reachable regions with high potential information gain. The selected frontier is then used as a navigation goal, and an A* planner computes a collision-free path to the target. To balance responsiveness and computational cost, replanning is performed at fixed intervals rather than at every timestep.

Reinforcement-learning agents

In addition to the classical baseline, two reinforcement learning agents are evaluated: a Deep Q-Network (DQN) agent and a Proximal Policy Optimization (PPO) agent. Both agents operate on the same discrete action space and receive identical observations, consisting of a local occupancy patch, the robot's heading, and normalized time information. The reward function encourages efficient exploration by penalizing time steps and collisions while providing a sparse positive reward upon successful victim detection. Training is performed in procedurally generated environments to expose the agents to diverse layouts while maintaining controlled experimental conditions.

Table 1: Algorithm configurations used in the experimental evaluation.

Parameter	**Frontier**	**DQN**	**PPO**
Action space	4 discrete actions	4 discrete actions	4 discrete actions
Observation type	Occupancy grid	Local patch (11×11)	Local patch (11×11)
Planning / update frequency	Every 5 steps	Every step	Every step
Learning rate	–	1×10^{-4}	3×10^{-4}
Discount factor (γ)	–	0.99	0.99
Batch size	–	32	64
Replay buffer size	–	100,000	–
Policy update steps	–	–	2048
Number of epochs	–	–	10
Collision penalty	–	-5	-5
Step penalty	–	-0.05	-0.05
Detection reward	–	$+100$	$+100$
Training budget (timesteps)	–	200,000	200,000

To ensure transparency and reproducibility, the main training and inference configurations used for the frontier-based method and the RL agents are summarized in Table 1. These parameters define the learning dynamics, planning behavior, and exploration incentives that shape each method's performance.

Although convolutional architectures are commonly used for spatial observations, a multi-layer perceptron (MLP) policy was selected in this study. This choice reflects the fact that the observation is provided as a flattened vector rather than a structured image tensor. Using an MLP avoids unnecessary architectural complexity and ensures that the policy network matches the structure of the input representation.

2.3 Experimental protocol

The experimental evaluation was conducted using a controlled and fully reproducible protocol designed to enable a fair comparison between the evaluated exploration methods. For each algorithm and experimental

condition, multiple independent episodes were executed in the same simulation environment, with only the exploration strategy varying across runs.

At the beginning of each episode, the environment was initialized using a deterministic random seed to ensure reproducibility while still allowing variation across maps and trials. The robot was then deployed in an initially unknown environment and allowed to explore autonomously until either the victim was detected or a predefined maximum number of timesteps was reached. Episodes terminated immediately upon successful detection, while unsuccessful runs were truncated at the time limit and recorded as failures.

To account for stochasticity in both the environment and the learning-based methods, each exploration strategy was evaluated across multiple procedurally generated maps, with several independent runs performed on each map. This design enabled the separation of variability due to map structure from variability due to random initialization or agent behavior. Reinforcement learning agents were evaluated using fixed, trained policies, and no learning occurred during evaluation.

During each episode, per-step state information and per-episode summary statistics were logged automatically by the simulator. These logs included detection time, path length, explored area, and success indicators, ensuring that all performance metrics were computed from a common data source. The same logging and termination criteria were applied uniformly across all methods, guaranteeing that performance differences reflect the exploration strategy rather than experimental artifacts.

2.4 Variables and measures

We manipulate a compact set of independent variables that are both meaningful for SAR performance and feasible to evaluate within the project scope. These variables are:

- Exploration method (primary): the algorithms compared (Frontier, DQN, PPO).
- Map complexity / scale (primary): combined factor representing grid size and obstacle density. Example levels: low (40×40, low density) and high (80×80, high density).
- Sensor noise (primary): measurement noise in range readings. Example levels: nominal ($\sigma = 0.2$) and noisy ($\sigma = 0.6$).
- Training budget (secondary): total timesteps used to train RL agents. Example levels: 100k/200k.
- Replication (secondary): number of independent maps and runs per map.

To evaluate the obtained results, for every episode we logged the following dependent variables:

- Time-to-find (T): timesteps to first detection (censored at $T_{\max}$).
- Coverage at detection (C): fraction of free cells observed at detection time.
- Path length (L): cumulative forward steps until detection.
- Success rate: boolean indicator if detection occurred before $T_{\max}$.

All metrics were aggregated per condition using median and inter-quartile range (IQR).

2.5 Statistical analysis

Because empirical distributions are skewed, primary summaries use median and inter-quartile range (IQR). Pairwise method comparisons use the Mann–Whitney U test (two-sided) with $\alpha = 0.05$; Bonferroni correction is applied for multiple comparisons. Effect sizes are reported using Cliff's delta; success-rate differences are tested with Fisher's exact test or two-proportion tests as appropriate. Analysis code uses NumPy, Pandas and SciPy (references and exact scripts are provided in the repository).

2.6 Reproducibility statement

All code, YAML configuration files and the Conda environment file are maintained in the project repository available at:
`https://github.com/baldassarreFr/SAR-Exploration-Algorithms`.

The repository includes (1) the exact commit hash used to generate results, (2) all configuration files, (3) sample commands to reproduce training and evaluation, and (4) trained checkpoints. A minimal reproduction script and the Conda environment file are included in the Appendix to ensure full experimental reproducibility.

2.7 Ethics and environmental impact

All experiments were conducted entirely in simulation, avoiding risks to people, infrastructure, or real robots. No personal data were collected, and all maps are either synthetic or publicly available. Ethical considerations focused on safety and reliability (ensuring that learned policies are not interpreted as deployable without further validation), transparency (providing an interpretable classical baseline), and responsible use of AI. Environmental impact was minimised by constraining training budgets, reporting compute resources, and favouring lightweight models. To support long-term sustainability, all code, configurations, and seeds will be released to enable reuse and reproducibility, thereby reducing duplicated computational effort across future work.

3 Results and Analysis

This section presents the experimental results obtained from the evaluation of frontier-based exploration and reinforcement learning–based exploration using DQN and PPO. All methods were evaluated under identical simulation conditions following the protocol described in Section 2. Performance is reported using the median and inter-quartile range (IQR), and unsuccessful episodes are treated as right-censored at the maximum episode length.

3.1 Time-to-Find Performance

Time-to-find is the primary metric used to assess exploration efficiency. Figure 2 summarizes time-to-find performance across exploration methods. The median values highlight typical detection speed, while the boxplots reveal variability and outliers across runs. In many unsuccessful reinforcement learning episodes, the time-to-find distribution collapses to the maximum episode length, indicating failure to locate the victim within the allowed time budget.

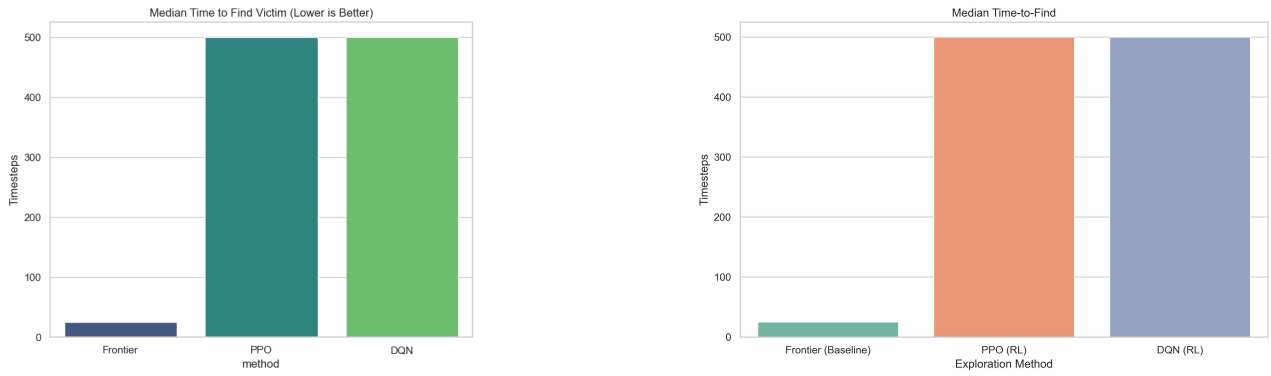


Figure 2: Time-to-find performance across exploration methods. *Left:* Median time-to-find with inter-quartile range. *Right:* Distribution of time-to-find values across successful episodes. Lower values indicate faster detection.

Reinforcement learning agents achieve very low median time-to-find values when detection occurs. In particular, PPO exhibits the lowest median time-to-find, followed closely by DQN, indicating that learned policies are capable of rapidly reaching the victim in favorable conditions. However, this behavior is observed only in a limited subset of runs, as reflected by the low success rates discussed in Section 3.4.

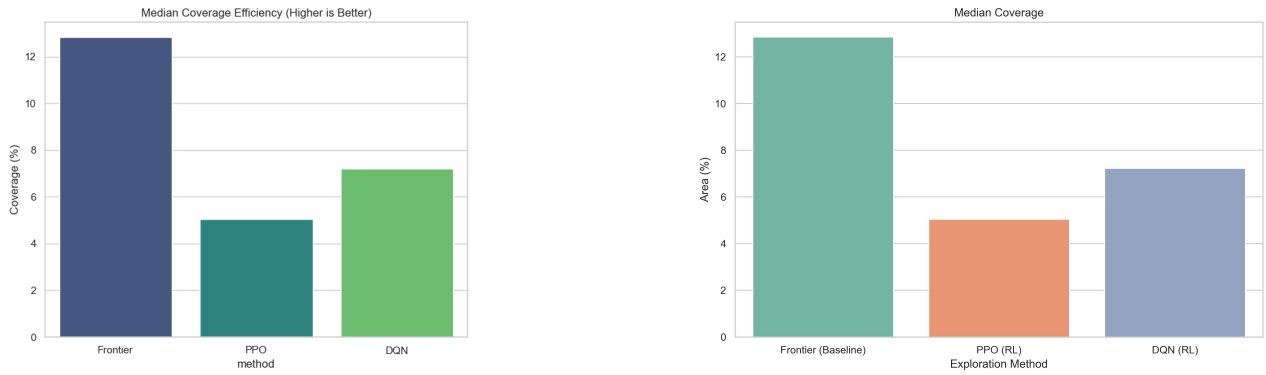


Figure 3: Coverage efficiency at the time of victim detection. *Left:* Median explored area percentage. *Right:* Distribution of coverage values across successful episodes. Higher values indicate more systematic exploration.

In contrast, frontier-based exploration displays a higher median time-to-find but with substantially greater consistency across runs. The inter-quartile range is wider than that of the RL agents, reflecting longer exploration trajectories, but detections occur reliably across a broad range of environments. This result highlights a trade-off between fast detections in isolated cases and robust performance across diverse conditions.

3.2 Coverage Efficiency at Detection

Coverage efficiency measures the proportion of the environment explored at the time of victim detection. Figure 3 illustrates both the median coverage efficiency and the distribution of explored area at the time of detection across methods.

Frontier-based exploration consistently achieves higher coverage at detection compared to both reinforcement learning agents. This indicates that detection typically occurs after a systematic expansion of explored space, which is characteristic of geometry-driven frontier selection. The relatively narrow distribution of coverage values further suggests stable exploration behavior across different maps.

Both DQN and PPO detect the victim after exploring a smaller fraction of the environment, reflecting their tendency to exploit local observations rather than performing global exploration. While this behavior can lead to very rapid detections in some cases, it also contributes to failure in more complex or unfamiliar layouts, where insufficient coverage prevents successful discovery. These differences in coverage efficiency are statistically significant across methods, supporting the conclusion that frontier-based exploration achieves more systematic coverage prior to detection.

3.3 Path Length Analysis

Path length captures the total distance traveled by the robot prior to detection and provides insight into navigation efficiency. Figure 4 shows the distribution of path lengths observed across successful episodes for each method.

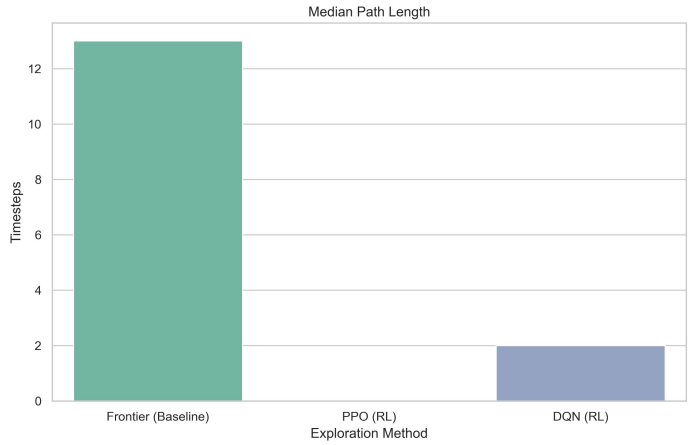


Figure 4: Distribution of path lengths until victim detection for each exploration method. Shorter paths indicate more directed navigation behavior.

The reinforcement learning agents exhibit very short path lengths in successful runs, consistent with their low time-to-find values. PPO in particular often reaches the victim with minimal movement, indicating highly directed behavior when a suitable trajectory is discovered early.

Frontier-based exploration produces longer but more consistent path lengths, corresponding to deliberate navigation toward frontier regions. The absence of extreme outliers suggests that the planner avoids unnecessary revisiting of explored areas and maintains coherent trajectories throughout exploration.

3.4 Success Rate and Robustness

Success rate reflects the proportion of evaluation episodes in which the victim is detected within the allowed time budget. Frontier-based exploration achieves a 100% success rate, successfully detecting the victim in all 30 evaluation episodes. This confirms its robustness to variations in map structure. In contrast, both reinforcement learning agents exhibit substantially lower success rates: PPO succeeded in only 10% of the runs, while DQN reached a success rate of 25%. Although they occasionally achieve extremely fast detections, they fail in a large fraction of runs, particularly in environments that differ from the training distribution. These failures are typically associated with insufficient environmental coverage or premature convergence to ineffective exploration behaviors.

3.5 Summary of Results

Overall, the results reveal a clear trade-off between efficiency and robustness. Reinforcement learning agents are capable of very fast victim detection in favorable scenarios, achieving minimal time-to-find and path length when successful. However, this efficiency comes at the cost of low success rates and reduced coverage, limiting their reliability in diverse environments. Frontier-based exploration, while slower on average, provides stable, predictable, and robust performance across all evaluated metrics, making it a strong baseline for search-and-rescue exploration tasks.

4 Discussion

This section interprets the experimental results in relation to the research questions and hypotheses defined in Section 1.2. The discussion focuses on the trade-offs between exploration efficiency, robustness, and generalization that emerge from the comparative evaluation.

4.1 Answering the Research Questions

Regarding **RQ1**, the results show that reinforcement learning agents achieve lower time-to-find values in successful runs, particularly PPO, which often reaches the victim very quickly. However, frontier-based exploration achieves more consistent detection times across runs, highlighting a trade-off between speed and reliability.

For **RQ2**, frontier-based exploration demonstrates higher coverage efficiency at detection. This indicates that the frontier method systematically explores the environment before detection, whereas reinforcement learning agents often detect the victim after exploring only a limited region of the map.

In response to **RQ3**, reinforcement learning agents produce shorter paths in successful episodes, reflecting highly directed navigation behavior. Frontier-based exploration, by contrast, follows longer but more structured paths driven by explicit planning toward unexplored regions.

Finally, **RQ4** is addressed through the observed success rates. Frontier-based exploration generalizes reliably across different environments, while reinforcement learning agents exhibit a pronounced performance drop, particularly in environments that differ from the training distribution.

4.2 Evaluation of Hypotheses

The experimental results partially support **H1**, as reinforcement learning agents achieve low time-to-find values in environments where they successfully detect the victim. However, the low success rates limit the practical relevance of this advantage.

H2 and **H3** are strongly supported by the results. Frontier-based exploration consistently outperforms reinforcement learning agents in terms of robustness, success rate, and stability across runs, particularly in diverse or unfamiliar environments.

H4 is also supported. Reinforcement learning agents tend to discover short paths in familiar scenarios but exhibit reduced coverage efficiency, which negatively impacts generalization and overall task success.

4.3 Why Reinforcement Learning Underperformed

The inferior performance of reinforcement learning agents can be attributed to several interacting factors. First, the sparse reward structure provides meaningful positive feedback only when the victim is detected. Given the limited training budget, agents rarely experience successful episodes during training, resulting in insufficient learning signals.

Second, the partially observable setting restricts the agent's perception to a local 11×11 window. Without an explicit memory mechanism, such as a recurrent architecture, the agents are unable to reason about previously visited regions, leading to inefficient exploration behavior. In practice, this manifests as random-walk patterns for DQN and overly conservative behavior for PPO, where the agent avoids movement to reduce collision penalties. It is worth noting that the choice of a memory-less MLP policy was deliberate. It serves to establish a baseline for reactive 'tabula rasa' agents, highlighting that without recurrent architectures (like LSTMs), end-to-end RL cannot yet compete with geometric methods in partially observable SAR tasks.

Finally, the use of held-out evaluation maps exposes a generalization gap. While the frontier-based method relies on geometric principles that transfer naturally across environments, the learned policies overfit to the statistical structure of the training maps and fail to adapt to unseen layouts.

5 Conclusion

The experimental results demonstrate that reinforcement learning agents did not outperform the frontier-based baseline under the evaluated conditions, leading to the rejection of Hypothesis H1. In contrast, Hypotheses H2 and H3 are strongly supported, as the frontier-based method exhibited superior robustness, consistency, and generalization performance on unseen environments.

These findings highlight an important methodological insight: in search-and-rescue scenarios where training data, compute resources, or prior environment knowledge are limited, classical geometric exploration algorithms provide a safer and more reliable baseline than tabula rasa deep reinforcement learning approaches. Future work may explore hybrid methods that combine the robustness of frontier-based planning with learned local navigation strategies.

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